Darren Zal

Researcher; Systems Designer; Software Engineer
Atlas Research Group
Victoria, BC / Salish Sea bioregion
GitHub: https://github.com/DarrenZal
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Research and Technical Focus

Darren Zal works on coordination infrastructure for plural human, organizational, and AI systems. His current work focuses on shared coordination substrates: memory, commitments, claims, evidence, provenance, permissions, signals, and boundary structures that shape how people and AI agents coordinate across contexts. He develops AI-assisted knowledge systems, graph-based research infrastructure, and governance tools for bioregional, Indigenous economic, ecological, and regenerative-finance settings.

His contribution to the Schmidt Sciences Trustworthy AI proposal is in substrate framing, coordination failure-mode analysis, topology/sheaf-informed evaluation ideas where useful, and practical testbed design for observable coordination health, evidence integrity, and repair pathways.

Current Roles and Projects

Atlas Research Group

Researcher; Systems Designer
Contributes to research on AI coordination infrastructure, substrate-level observability, membranes, shared memory, and formal/practical evaluation surfaces for multi-agent systems. Current work emphasizes how coordination media shape behavior before individual agent outputs appear, and how governance-relevant signals can remain visible without collapsing plural contexts into a single central model.

Open Civics / Civic Intelligence Engine

Delegate; Project Contributor
Contributes to Open Civics as a delegate and is beginning work on the Civic Intelligence Engine, an OpenCivics Labs / Better Governance Institute project for AI-assisted constituent engagement, civic sensemaking, and public-facing visualization of community voice. Relevant work includes public-interest AI, participatory feedback systems, AI-facilitated dialogue, structured analysis of public input, and governance-facing interfaces that make collective signal legible without replacing human political judgment.

Kwaxala

Co-Founder; Systems Architect
Designs technical and strategic infrastructure for forest protection and regeneration. Work includes architecture for the Living Forest investment system, aligning ecosystem stewardship, Indigenous guardianship, and long-term financing structures for legally protected living forests.

Indigenomics AI

Researcher and Developer
Builds AI-assisted knowledge and research infrastructure for Indigenous economic systems, including worldview-sensitive knowledge modeling, source-grounded synthesis, graph-based inquiry, and consent-aware evidence practices. Relevant work explores how data sovereignty, provenance, OCAP-aligned participation, and plural worldviews can be represented in AI-supported coordination systems without flattening local authority.

Regen AI / Regen Network

Researcher and Developer
Works on AI and knowledge-infrastructure for regenerative claims, ecological evidence, knowledge graphs, commitment pools, and coordination workflows. Relevant themes include claims/evidence lifecycle design, provenance-bearing AI systems, federated knowledge graphs, and tooling for ecological and regenerative-finance coordination.

Symbiocene Labs / Gaia AI

Co-Founder; AI Developer
Develops AI and coordination infrastructure for regenerative systems transformation. Gaia AI is a Symbiocene Labs brand/project focused on interoperable AI agents and human-AI collaboration for collective sensemaking, ecological intelligence, and systemic problem solving.

Ecoscene Oasis

Co-Founder; Lead Developer
Leads software and systems development for a platform supporting regenerative education, coordination, and community knowledge practices. Work spans product architecture, AI-enabled workflows, and infrastructure for collective sensemaking and action.

Salish Sea Dreaming

Project Lead / Systems Designer
Develops bioregional cultural and knowledge infrastructure for dream, story, place, and collective imagination in the Salish Sea. Relevant work connects participatory knowledge contribution, field mapping, narrative evidence, and shared meaning-making across communities.

Regenerate Cascadia / Bioregional Coordination

Contributor
Contributes to bioregional coordination research and infrastructure, including knowledge commons, landscape-hub coordination, flow funding, commitment mapping, and regenerative economic protocols for the Cascadia bioregion.

Spore / Agent Commons

Creator and Maintainer
Develops a coordination grammar and protocol family for plural, sovereign coordination across humans, AI agents, teams, organizations, and federations. Spore defines structural primitives such as fields, holons, and membranes, and coordination operations such as intents, commitments, evidence, signals, and reviewable authority. The work is directly relevant to substrate-level coordination: it treats shared memory, permissions, commitments, and boundary operations as the medium through which coordination becomes observable and governable.

Prior Professional Experience

Longtail Financial

Token Engineer
Researched and simulated token-economic systems for Web3 and regenerative finance. Co-created and mentored an eight-week Regenerative Finance Bootcamp for builders designing practical regenerative economic models.

Grassroots Economics

Software Developer, Freelance
Contributed open-source software, metrics dashboards, and data-analytics tools supporting community currency systems and grassroots economic resilience.

Fast Enterprises

Implementation Consultant / Software Developer
Delivered public-sector revenue and benefits software for provincial and state governments, including British Columbia, Alaska, and Vermont. Work included requirements gathering, scope definition, live demos, deployments, custom software extensions, complex reporting, SQL development, and batch/data-warehouse optimization.

Selected Research and Infrastructure Themes

- Multi-agent and human-AI coordination substrates
- Shared memory, provenance, evidence, permissions, and commitments
- Graph-based knowledge systems and retrieval infrastructure
- Coordination membranes, consent, authorization, and revocation
- Bioregional knowledge commons and federated governance
- Public-interest AI, civic intelligence, and participatory feedback systems
- Indigenous data sovereignty and worldview-sensitive knowledge modeling
- Regenerative finance, commitment economies, and ecological claims infrastructure
- AI-assisted sensemaking, research synthesis, and decision support

Technical Skills

Python; TypeScript; JavaScript; Svelte/SvelteKit; HTML/CSS; SQL; Solidity; VB.NET; AI-assisted software development; systems architecture; full-stack product engineering; Model Context Protocol (MCP) server and tool-interface development; agentic workflow design; retrieval-augmented generation and GraphRAG; knowledge-graph and semantic-search systems; provenance, claims, evidence, and consent-aware data modeling; public-interest AI and civic intelligence tooling; data ingestion, ETL, dashboards, and interactive visualization; API integrations; local-first knowledge systems; public-sector software delivery; token engineering and regenerative-finance modeling; open-source development.

Education

Bachelor of Science, Physics and Computer Science
McGill University, 2014